

The steeper $1/f$ slope of MEG power spectrum is associated with stronger gamma synchrony, measured via inter-trial phase coherence (ITPC) in the 40-Hz auditory steady-state response

Introduction

Coordinated activity of neurons and robust mechanism to response to external stimuli are crucial for brain function and its role in organism survival. Neural phase-locking represents a fundamental process where local neuronal ensembles synchronize their oscillatory activity to external rhythmic stimuli, enabling precise temporal coordination of neural responses. Among the various frequency bands of neural oscillations, gamma-band activity (30-100 Hz) serves as a particularly important mechanism for local circuit coordination and stimulus-locked information processing (Fries 2009). However, traditional approaches to studying neural phase-locking often examine oscillatory activity in isolation, without considering how it relates to other neural markers that may reflect similar underlying physiological mechanisms.

Understanding the relationship between different electrophysiological measures is becoming increasingly important as researchers seek biomarkers for brain health and its dysfunctions. Among the measures that have emerged as potentially important indices of brain activity (Pellegrino et al. 2024) are gamma-band synchrony (estimated via the ability of neural populations to maintain consistent phase relationships to external stimuli across repeated presentations) and the $1/f$ component of neural power spectra (spectral slope). Both may reflect fundamental properties of local neural circuits, particularly the balance between excitation and inhibition (E/I balance), yet they are rarely examined together within the same experimental framework.

Neural oscillations and gamma synchrony

Neuronal oscillations are rhythmic changes in neuronal ensembles between high and low excitability states (Schroeder and Lakatos 2009). They shape sensory, motor and cognitive processes (Henao et al. 2020) and are thought to be the building blocks of fast-scale information processing in the brain serving as a mechanism for synchronising neural activity across distributed regions (Womelsdorf et al. 2007). During oscillations, a probability of neuron discharge is modulated by local extracellular and membrane potentials that in turn are reflected by the phase (Henao et al. 2020). This might create a “window of integration”, where stimuli falling within the high excitability part of the cycle might be integrated, especially through gamma oscillation (Fries 2009).

Gamma oscillations occupy a unique position in this oscillatory hierarchy due to their role in local circuit dynamics and stimulus processing. They are primarily generated by parvalbumin-expressing inhibitory interneurons through precisely timed interactions with pyramidal neurons (Bartos, Vida, and Jonas 2007). As Fries (Fries 2009, 2015) argues, gamma-band activity

enables cortical computation through precise temporal coordination that allows for both selectivity and invariance of neuronal responses. This coordination enables functional segmentation of neural processing, where local populations can phase-lock to relevant stimuli while maintaining independence from irrelevant inputs. The capacity for gamma synchrony - the ability of neural populations to consistently align their oscillatory activity to external rhythmic stimuli across multiple stimulus presentations (Schuler et al. 2022) - is critical for normal brain function. Alterations in this phase-locking capacity often indicate pathological conditions (Pellegrino, Arcara, Cortese, et al. 2019). Changes in local gamma phase-locking have been consistently observed in various neuropsychiatric disorders, including schizophrenia, autism spectrum disorders, and bipolar disorder, making it an important biomarker for local circuit dysfunction (Parciauskaite, Bjekic, and Griskova-Bulanova 2021).

Auditory Steady-State Responses (ASSR) as a tool for studying neural synchrony

The auditory steady-state response (ASSR) represents the ability of neural populations to phase-lock their oscillatory activity to the frequency and phase of repetitive auditory stimuli and provides a robust experimental paradigm for investigating local gamma synchrony in controlled laboratory conditions. Lakatos, Gross, and Thut (2019) defined entrainment as phase locking resulting from (predominantly) unidirectional coupling.

The 40-Hz ASSR has emerged as a particularly valuable tool for studying local gamma phase-locking. Tada et al. (2020) indicate that the 40-Hz ASSR is generated primarily through synaptic interactions between parvalbumin-positive GABAergic interneurons and pyramidal neurons, making it a direct probe of the local neural circuits that regulate E/I balance. This response is primarily generated in the medial region of the primary auditory cortex, with additional contributions from thalamic and corticothalamic sources, and recent evidence suggests significant involvement of prefrontal cortex in gamma-range ASSR generation (Parciauskaite, Bjekic, and Griskova-Bulanova 2021).

The 40-Hz ASSR shows several characteristics that make it particularly useful for research into local phase-locking mechanisms. It demonstrates high test-retest reliability and can be measured consistently across different species, facilitating translational research (Tan, Gross, and Uhlhaas 2015). The response can be elicited reliably across various populations, though it shows both inter- and intra-subject variability that may reflect meaningful individual differences in local neural circuit properties. Importantly, ASSR phase-locking deficits have been documented as early indicators of brain dysfunction, appearing in at-risk populations before the onset of clinical symptoms (Tada et al. 2020). Research by Parciauskaite and colleagues (Parciauskaite, Bjekic, and Griskova-Bulanova 2021) demonstrates that gamma-range ASSR phase-locking ability is related to cognitive flexibility and reasoning abilities in healthy participants, and to memory and processing speed in patients with schizophrenia. These findings suggest that individual differences in gamma-range ASSR reflect important aspects of local circuit functioning, including the precision of stimulus-locked neural responses and the stability of local gamma-generating networks.

Aperiodic neural activity and the 1/f slope

Complementary to rhythmic neural activity, electrophysiological signals contain a prominent component that follows a 1/f power law. This component has a characteristic shape where power decreases exponentially with increasing frequency, creating a linear relationship when plotted in log-log space (Donoghue et al. 2020). The 1/f component (slope) can be characterized by two main parameters: an offset that reflects broadband power across all frequencies, and an exponent (slope) that determines the rate of power decay across frequencies.

Traditionally, this aperiodic activity was considered neural “noise” and was either ignored or statistically removed from analyses. However, mounting evidence demonstrates that the aperiodic component contains meaningful physiological information (Donoghue et al. 2020). The aperiodic offset correlates with neuronal population spiking activity and the BOLD signal from functional MRI, suggesting it reflects the overall level of neural activity. More intriguingly, the aperiodic exponent has been linked to the integration of underlying synaptic currents, which have stereotyped temporal dynamics that naturally give rise to the 1/f-like nature of neural power spectra.

Gao and colleagues (Gao, Peterson, and Voytek 2017) propose that the aperiodic exponent reflects the balance between excitatory and inhibitory synaptic currents. Excitatory currents, particularly those mediated by AMPA receptors, have relatively fast time constants and contribute to flatter power spectra (lower exponents). In contrast, inhibitory GABA currents have slower time constants and contribute to steeper power spectra (higher exponents). This means that the exponent will be lower when excitation dominates and higher when inhibition is stronger, providing a potential non-invasive marker of E/I balance.

The clinical and developmental relevance of the aperiodic exponent is becoming increasingly apparent. Changes in spectral slope have been documented across the lifespan, with systematic alterations observed in aging (Voytek et al. 2015) and various neuropsychiatric conditions. The slope shows trait-like properties that are stable within individuals but vary meaningfully between individuals and groups, suggesting it may serve as a biomarker for brain function and dysfunction (Ahmad et al. 2022).

Potential relationships between 1/f slope and gamma phase-locking

The relationship between the aperiodic exponent and local gamma synchrony could arise through multiple, non-mutually exclusive mechanisms. The most straightforward connection is biological: both measures may reflect shared underlying processes related to local E/I balance. If gamma synchrony depends on the precise timing and strength of inhibitory interneuron function, and the aperiodic exponent reflects the relative strength of excitatory versus inhibitory currents within local circuits, then systematic relationships between these measures would be expected across individuals and brain regions.

Supporting this biological connection, both ASSR phase-locking deficits and changes in aperiodic slope co-occur in similar neuropsychiatric conditions. For example, schizophrenia is characterized by both reduced 40-Hz ASSR phase-locking and altered 1/f slopes (Martínez-Cañada et al. 2023). Similarly, autism spectrum disorders show both local gamma phase-locking deficits and changes in aperiodic activity. These parallel alterations suggest that both measures may be influenced by common pathophysiological mechanisms, particularly those involving GABAergic interneuron function and NMDA receptor-mediated excitation within local circuits.

However, the relationship between these measures may also have important methodological components. Recent work by Samaha and Cohen (Samaha and Cohen 2022) demonstrates that the slope of the power spectrum can systematically confound estimation of oscillatory properties, including instantaneous frequency and phase measurements. Since inter-trial phase coherence (ITPC) depends critically on consistent phase relationships across trials within a given brain region, systematic variations in spectral slope could introduce bias in local phase-locking estimates that is independent of any true biological relationship.

This methodological concern is particularly relevant for ITPC measurements when signals lack clear oscillatory peaks above the aperiodic background, or when signal-to-noise ratios are low due to dominant 1/f components (Zrenner et al. 2020). In such cases, the phase values obtained from standard time-frequency analysis methods may become unreliable, potentially leading to spurious correlations between 1/f slope and apparent changes in measures of synchrony (van Diepen and Mazaheri 2018). The concept of “phase” is inherently linked to rhythmic oscillations, and when clear oscillatory activity is difficult to distinguish from 1/f background activity, phase-based measures like ITPC may be systematically biased by the spectral slope (Wodeyar et al. 2023).

Current study

This study investigates the relationship between the aperiodic spectral exponent and local gamma phase-locking as measured by ITPC during 40-Hz ASSR in healthy participants. We analyzed MEG data from 15 participants during auditory stimulation.

Our analytical strategy utilizes Bayesian multilevel modeling to examine how spectral slope might be related to ITPC measured across brain regions while appropriately handling the multiple comparisons inherent in whole-brain analysis. This approach allows us to take into account region-specific effects while borrowing strength across regions through partial pooling, providing more robust and interpretable results than traditional mass-univariate approaches.

Understanding the relationship between these two putative biomarkers of local neural circuit function has important implications for their use in clinical and research contexts. If they reflect shared biological mechanisms, their combined examination may provide more sensitive and specific markers of local circuit integrity than either measure alone.

Definitive testing of biological relationship is not easy, however, we can partially test methodological confound hypothesis with simulations of signals with varying parameters. The slope-ITPC relationship may partially reflect methodological interdependencies in phase estimation, where steeper slopes (indicating stronger high-frequency attenuation) systematically bias ITPC measurements toward lower values by reducing the signal-to-noise ratio of oscillatory components relative to aperiodic background.

Participants and experimental design

This study analyzed a magnetoencephalography (MEG) dataset previously described by Pellegrino, Arcara, Di Pino, et al. (2019). The experimental procedures were approved by the Ethics Committee of the Province of Venice, and all participants provided written informed consent before participation.

Fifteen healthy, right-handed volunteers participated (12 females, 3 males; mean age = 28.8 ± 3 years, age range: 20-50 years). Participants were screened using the following exclusion criteria: (1) hearing impairment, (2) current use of medications affecting the central nervous system, (3) history of neuropsychiatric disorders, and (4) any other medical condition that might affect brain function.

The experiment used a within-subjects design to investigate cortical gamma-band synchronization in response to 40 Hz amplitude-modulated auditory stimuli. As shown in Figure 1, the protocol comprised two different sessions of bi-hemispheric tDCS stimulation (sham and real) conducted at least 20 hours apart indicated by condition (T): REAL or SHAM. The order of the tDCS stimulation (sham and real) was counterbalanced across subjects. Stimulation was applied to the left and right pericentral regions Figure 1 using standard protocols. The anode was positioned over the left hemisphere and the cathode over the right hemisphere, targeting auditory cortical regions. Real stimulation lasted 20 min with 20 s of fade-in and fade-out. Sham stimulation was performed with the same settings, but electrical current was delivered only for 20 s at the beginning and at the end of the tDCS session.

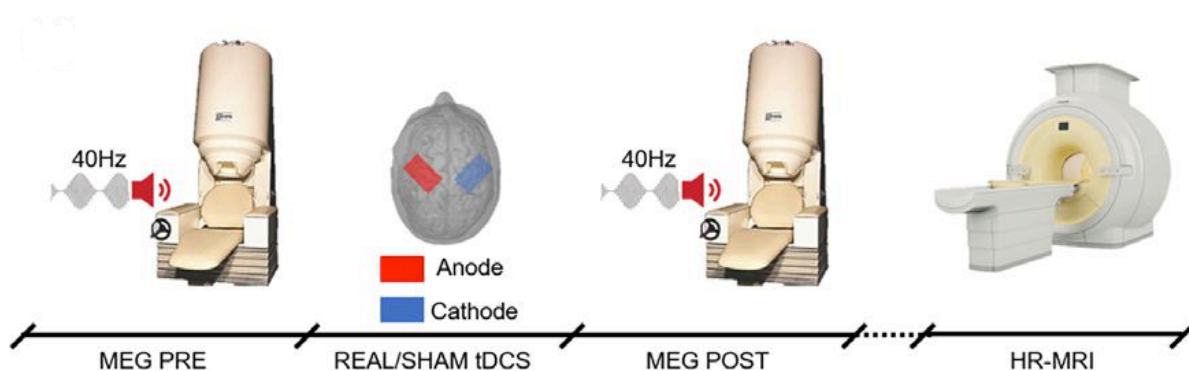
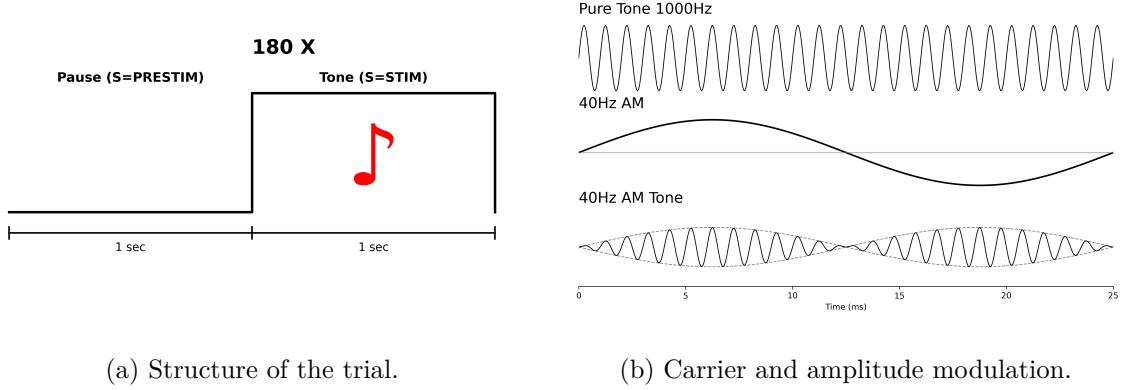


Figure 1: Design of the original experiment. Source: (Pellegrino, Arcara, Di Pino, et al. 2019)

Each session included two blocks corresponding to conditions related to the time (P) of applying tDCS stimulation: PRE (before tDCS) and POST (after tDCS). Figure 2a shows the structure of a paradigm used to induce 40 Hz synchrony during a single block. Each block consisted of 180 repetitions lasting 2 s each. Each trial included 1 s of pause (S=PRESTIM condition) followed by a 40 Hz amplitude-modulated (AM) tone (S=STIM condition), so that the stimulus onset asynchrony was 2 s. Each block lasted a total of 6 minutes (180 trials $\times$ 2 seconds). Figure 2b illustrates the auditory stimulus, an amplitude modulated tone with the following parameters: carrier frequency = 1,000 Hz and amplitude modulating frequency = 40 Hz.



(a) Structure of the trial.

(b) Carrier and amplitude modulation.

Figure 2: Structure of auditory stimulation. Adapted from Pellegrino, Arcara, Di Pino, et al. (2019).

MEG Data acquisition and preprocessing

MEG scans were performed at the MEGLab of the IRCCS San Camillo Hospital in Venice in a shielded room with a CTF-MEG system equipped with 275 gradiometers. As outlined in (Pellegrino, Arcara, Di Pino, et al. 2019) MEG data preprocessing included: third-order spatial gradient noise cancellation, downsampling to 600 Hz, signal space-separation. Heartbeat and eye-movement artefacts were cleaned with the SSP approach. MEG data was cut in 3 s epochs ranging from -1.5 s to $+1.5$ s around the 40 Hz sound onset. Epochs were visually inspected and those affected by artefacts were manually rejected. The available MRI scans were segmented and processed through Freesurfer software. The surface was divided into 15,000 vertices imported into Brainstorm toolbox and downsampled into 8,000 vertices. The coregistration between functional MEG data and anatomical MRI data was performed via the Brainstorm toolbox.

Data was bandpass filtered between 2 and 80 Hz and then notch filtered for 50~Hz. The inverse problem for source estimation was solved with the whitened and depth-weighted linear L2-minimum norm estimate, with the dipole orientation constrained to be normal to the cortical surface. A common imaging kernel was computed and then applied to obtain single

epoch cortical reconstructions. Data was projected into regions using Destrieux et al. (2010) atlas and exported from Brainstorm into MNE Python.

PSD and Aperiodic Parametrization

For every trial, the 1s second fragment before sound stimulation and 1s fragment during the sound stimulation was extracted. For those intervals power spectral density was computed using MNE Python using `welch` method and the median was taken across trials in a block (as is more robust than just averaging). Then the slope and exponent of the spectrum were computed for each region of interest using `specparam` toolbox (Donoghue et al. 2020).

In the algorithm the power spectrum PSD is modeled as a combination of the aperiodic component AP and a sum of N oscillatory peaks modeled as a Gaussians:

$$P = AP + \sum_{n=0}^N G_n.$$

The component $AP(f)$ for frequency f can expressed by the formula:

$$AP(f) = b - \log(k + f^\chi)$$

where b is the broadband offset, χ is the exponent and k is the knee parameter. The exponent χ is related to the negative slope of the power spectrum in log-log space, that is, the inverse relationship between power and frequency; The offset b describe the uniform shift in power across frequencies. The frequency range on which the algorithm was fitted span interval $[3, 45]$. The parameters with which data were analysed are the following hyperparameters : `peak_width_limits=[1, 8]`, `min_peak_height=0.15`, `peak_threshold=2.0`.

Estimation of synchrony via inter-trial phase coherence (ITPC)

Spectral modulations and phase consistency of ASSR in MEG and EEG data are highly reproducible, providing support for other research (Tan, Gross, and Uhlhaas 2015; Pellegrino, Arcara, Di Pino, et al. 2019). ASSR are usually investigated by looking at the inter-trial phase coherence (ITPC). The inter-trial phase coherence (ITPC) is defined using the length of the mean phase angle,

$$\text{ITPC} = \left| \frac{1}{N} \sum_{k=1}^N e^{i \cdot \phi_k} \right|,$$

where N represents the number of trials, $e^{i \cdot \phi_k}$ is complex polar representation the phase angle ϕ of the trial k . ITPC is an measure of phase consistency over trials, can take values between 0 and 1. The higher is ITPC, the higher is the 40 Hz gamma synchronization

The ITPC was computed for 1s second pre-stimulus period and 1s stimulation period via Time-Frequency Representation (TFR) using Morlet wavelets approach implemented in MNE Python (1.9) in the 39-41 Hz gamma band for each epoch.

Values of ITPC were extracted in the period of -500ms to -200ms in the period before sound stimulation starts and 300 ms to 700 ms during the sound stimulation, counting on the time of brain reaching steady-state (Pellegrino, Arcara, Di Pino, et al. (2019)) and averaged in those periods.

Mapping of value distributions of variables of interest

To examine potential differences in inter-trial phase coherence (ITPC) across experimental conditions, we conducted a Bayesian multilevel analysis using the brms package (Bürkner, 2017) in R (R version 4.5.1 and brms version 2.22).

First part we used data from the PRE condition

We looked into simple model where and how much ASSR change ITPC. My assumption is that the effect of sound stimulation will be dependent both from the Subject and roi (Barr et al. (2013) argues for maximal structure of model being conservative)

$$ITPC \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

Similarly we do use similar structure for parameters of the slope - exponent and offset

$$exponent \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

$$offset \sim 1 + S + (1 + S|ROI) + (1 + S|Subject)$$

We decided to fit sigma in the models as well to capture the uncertainty more precisely - Different subjects may not only differ in their mean responses but also in their consistency and different ROIs might have different signal-to-noise ratios.

Main Analysis

A hierarchical Bayesian model was implemented using the brms package to assess the relationship between ITPC and spectral exponent. The model was specified as:

$$itpc \sim 1 + exponent_z + (1 + exponent_z|roi) + (1 + exponent_z|subject)$$

This formula represents ITPC as a function of the standardized spectral exponent (exponent_z), while accounting for random intercepts and slopes for both brain regions (roi) and

subjects. We employed Student’s t-distribution as the likelihood function to accommodate potential outliers in the data. The model was fitted using Markov Chain Monte Carlo sampling with 4 chains, 4000 iterations per chain, and a 1000-iteration warm-up period. To improve convergence, we set the `adapt_delta` parameter to 0.95.

To interpret region-specific effects, we extracted random effects for each ROI using the `extract_region_effects` function. This allowed us to calculate:

Base ITPC values (intercept) for each brain region
 The effect of spectral exponent on ITPC in each region
 Posterior probabilities (P+) indicating the likelihood of positive effects
 95% credible intervals for parameter estimates

We calculated predicted ITPC values at different levels of the spectral exponent (± 1 standard deviation from the mean) to quantify effect sizes across regions. Brain regions were then ranked based on both the magnitude of the spectral exponent effect and the posterior probability of the effect being positive or negative. Results were visualized using brain plots to illustrate the spatial distribution of effects.

ITPC Comparison Between Experimental Conditions including tDCS

We constructed a hierarchical Bayesian model with ITPC as the dependent variable. Fixed effects included the main effects of measurement time point (P: pre vs. post), stimulation type (T: tDCS vs. sham), and their interaction (P×T). The model incorporated subject-specific and ROI-specific random intercepts and slopes for all fixed effects, allowing for individual variation in both baseline ITPC values and responses to the experimental manipulations. The model specification was:

$$itpc \sim 1 + P * T + (1 + P * T|roi) + (1 + P * T|subject)$$

We used a Student’s t-distribution as the likelihood function to accommodate potential outliers in the data. The model was fitted using four Markov Chain Monte Carlo (MCMC) chains with 4,000 iterations each (including 1,000 warmup iterations), resulting in 12,000 post-warmup samples for posterior inference. To improve convergence, we set the `adapt_delta` parameter to 0.95.

For hypothesis testing, we examined the posterior distributions of the regression coefficients, focusing on the main effects of time point and stimulation type, as well as their interaction. We calculated probability values (P+) for each ROI-specific parameter, representing the probability of a positive effect when the mean was positive, or a negative effect when the mean was negative. Effects were considered to show strong evidence when P+ exceeded 0.95 (strong evidence for a positive effect) or fell below 0.05 (strong evidence for a negative effect).

Simulations

Descriptive Statistics and Data Quality

Before examining the relationship between spectral exponent and ITPC, we first verified that our experimental manipulations (tDCS stimulation) did not systematically alter ITPC measurements, allowing us to pool data across conditions for our primary analysis

Results

ITPC

First we tested if the conditions PRE in real and sham conditions aren't different. No region has P + values bigger than 0.90 or smaller than 0.10 so we concluded that these conditions aren't different.

Simple visualization of the results ITPC results are presented in the Figure 3.

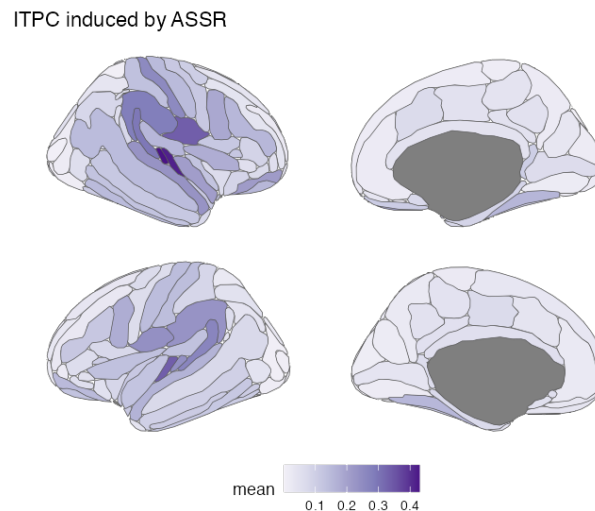


Figure 3: Z-normalized values of ITPC (for PRE Sham and PRE real conditions)

The region with the biggest response values were The $P+$ (Part of posterior distribution that is bigger than 0) is presented in the table

We also computed the map of difference Pre sham/real tDCS effects and then the Real versus Sham stimulation were contrasted (computed the difference) using Bayesian Multilevel Modelling, but no region has P + suggesting effect of TDCS.

Specparam parameters

We tested effects of stimulation on exponent and we didnt find any difference. By meaningful difference we count those with $P > 0.975$ or 0.025 . Below we present exponent values as mapped per region

Mean values of parameters describing the 1/f part of spectral power across different regions (right hemisphere).

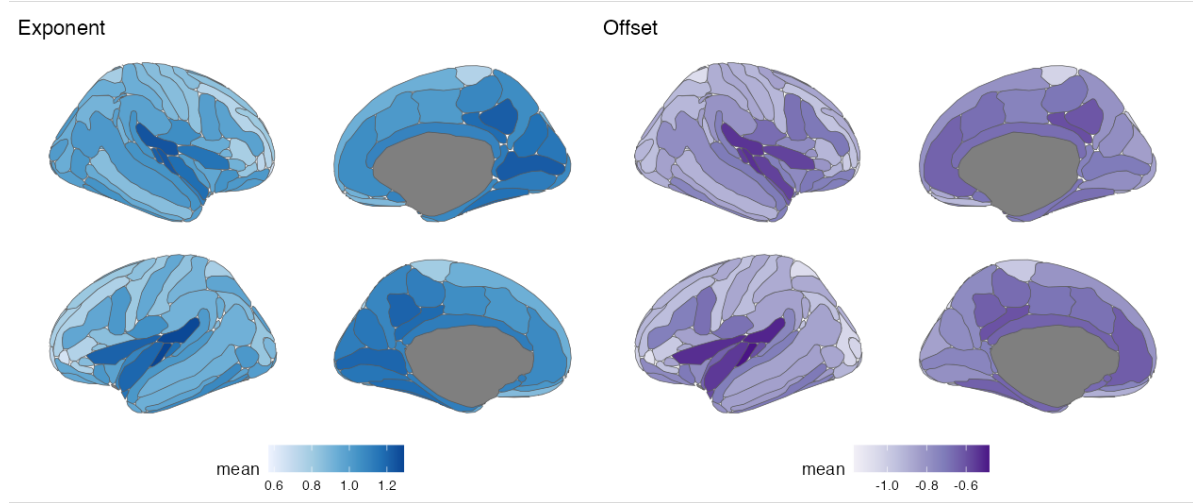


Figure 4: Mean values of exponent and offset averaged for the trials before tDCS stimulation and before sound stimulation

Evidence of Three Explanatory Models for the ITPC

- The exponent and the offset of the spectrum are different across the brain (Box 1); The parametrization of the slope of the aperiodic component (exponent) might be affected by neural entrainment and increased power in gamma frequency
- Strong effect of the stimulus (S) on the ITPC (contrast baseline-stimulus), in areas like the auditory regions, but also in temporal lobe, inferior part of primary sensory-motor regions and inferior frontal regions; negative values indicate higher ITPC during stimulation.
- Model with two explanatory variables

$$ITP \sim Exponent + Stimulus$$

has strong evidence in the same regions, both when considering exponent and stimulus as explanatory variables, with the effects having opposite signs, on the same regions.

- No particular evidence was found for the model $\text{ITPC} \sim \text{Exponent} + \text{Stimulus}$; namely the stimulation did not influence the exponent per se.
- No significant evidence was found for any of the models involving the Offset either as a predicted variable, and as a predictor, suggesting a marginal role for it in the auditory entrainment.

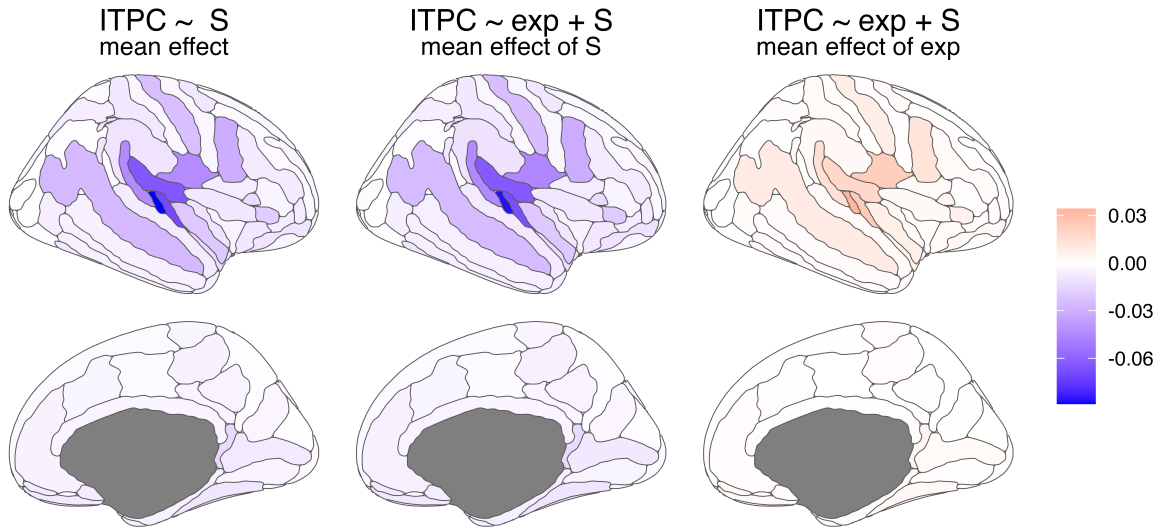


Figure 5: Positive values indicate stronger modulation at baseline, negative values indicate stronger modulation during stimulation.

ITPC including tDCS

Additionally, we extracted the ROI-specific coefficients to examine regional variation in the effects. This allowed us to identify potential regions showing differential responses to the stimulation conditions. The preliminary analysis revealed no significant differences in ITPC between pre- and post-stimulation timepoints, nor between tDCS and sham conditions. The interaction between timepoint and stimulation type also showed no significant effect. These findings suggest that ITPC measurements can be appropriately pooled across these conditions in subsequent analyses, as the experimental manipulations did not substantially alter the phase coherence measures.

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